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From colour target to ICC

InkProf - When colours have to be right

Source version 1.1.0, released 2026-10-12. Document updated 2026-10-13. Base MATLAB R2025b and Python 3.13 tested on macOS. Windows and Linux are untested; the measurement bridge uses POSIX. Install runtimes and ArgyllCMS separately. Start with setupInkProf(CheckPython=true); startInkProfApp(); from the source folder.

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One project, one source of settings

Use New project once, then Open project to resume. Project details has Project and materials, Printing settings, Profiling and Target paper tabs. Record the user, printer, paper name/finish, ink and dye/pigment/mixed/unknown type, printer coating and its settings, print application, media/driver/quality/colour management, drying hours, profile name, D50 colour-data choices, FWA and paper/sled limits. Unknown values remain unknown. Recipe fields managed here are read-only in the recipe dialog. FWA changes made later are reflected in the project. Changing relevant inputs invalidates dependent results, preserving earlier evidence.

Rename through Project details to rename the project folder too. External folder renaming is detected for confirmation. Move the entire project folder between computers; locally discovered Python and Argyll paths belong to the computer, not the portable project. Verify project checks recorded file hashes. Close app after saving/completing the active operation; reopen the same project after drying.

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The 19 rows

Row	Action	What it creates or decides
1	Create/import RGB target	RGB definition and generation metadata
2	Save TIFF16	Matching TIFF16, TI2, layout and manifest; print separately
3	Measure/select revision	Instrument readings and saved JSON/TI3 revision
4	Review and accept	Explicit acceptance of the selected revision
5	B1: lock inputs	Immutable, checked profiling input snapshot
6	B2: recipe	Manual build recipe; project settings are reused
7	B3: build ICC / automatic iteration	ICC candidate, build arguments and job log
8	Fit, Grid and C1	Model fit, transformations and numerical checks
9	C2: save verification TIFF16	Profile-applied verification target
10	Measure verification	Saved verification revision

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The 19 rows — continued

Row	Action	What it creates or decides
11	C3: compare model and print	Desired vs measured and predicted vs measured errors
12	Review feedback	Recorded feedback/decision, not merely opening a window
13	Approve intended use	Explicit scope, notes and confirmation tied to current evidence
14	Save ICC and certificate	Delivery copies plus PDF/HTML measurement certificate
15	Save refinement TIFF16	Added fitting patches, development checks and controls
16	Measure refinement	Reviewable additional measurements
17	Build next iteration	Combined training set and a new ICC candidate
18	Compare previous/current ICC	Shared RGB/Lab comparisons, independent of original meshes
19	Save without print verification	Numerically checked later iteration and explicitly limited certificate

Ready: Required inputs are available. Select the step and Run selected step to start.
Complete: The step’s result or decision is saved and the app’s requirements for that step are met. This does not itself mean approved print quality.
Locked: The step cannot start yet. Read its description and first complete the prerequisite steps.
Stale: An earlier result needs updating because its inputs or settings have changed. Repeat the affected steps.
Use the dialog’s intended save or acceptance button. Simply opening and closing the dialog does not complete the step.
Refresh reloads saved status; it does not save edits still open in a dialog. Finish or cancel the active dialog before starting another operation.

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Targets, paper and measurement

TIFF16 files are saved, not sent to the printer. Copy TIFF16 for printing copies the selected target; retain matching TI2/layout data in the project. Use the documented colour-management path: base/refinement device RGB targets must not receive an unintended additional ICC conversion; C2 already contains the intended profile transform. Print under the recorded conditions and allow drying. Row labels are on both sides; compact layouts reserve a footer band.

Target paper suggests cut sheets A5 through A3+ or a roll/custom size according to patch count and equipment limits. Defaults 320 mm scan width and 370 mm length describe one sled, not a universal limit; edit the JSON-backed Project details values and review suggestions before saving. Smaller iterations can therefore use smaller sheets.

Measure through the app, review instrument prompts and allow calibration. Instrument model and serial are captured from output when available, otherwise remain unknown. A remeasured patch is a candidate until accepted/saved as a new revision. Select that revision and repeat the dependent review/checks. Absolute Lab and change in Lab are separate quantities. See instrument-startup and chart-measurement guides for reconnect/retry behaviour.

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Refinement and shadows

Use image colours or C3 feedback to propose more patches. An image's embedded ICC is used; if absent, a warning precedes the sRGB fallback so you can choose another image. Estimated-error filtering defaults to ΔE00 > 5. These are predictions, not measured errors.

Unmeasured C2 colours can be included with the new target. C2 colour/gray/challenge patches train the next iteration; repeated and paper-white patches retain control roles. Once used for fitting they are not independent validation of that next ICC.

For Matte paper, the configurable InkProf shadow strategy defaults to targen emphasis 2, colprof grid emphasis 1.3 and up to 48 additional dark fitting patches per iteration. Initial targets redistribute their selected budget; iteration targets add dark patches without removing selected image/C3 colours. Selection uses the current ICC's lower quarter of its black-to-white L* range and excludes RGB16 duplicates. Actual counts and the source ICC hash are saved. This is not an official Argyll matte preset, and improved print accuracy requires measurement. Existing targets are unchanged.

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Interpret and deliver

Fit error against training data, numerical consistency and physical print verification answer different questions. Large desired-colour errors with small model-vs-print errors may reflect gamut limits; this alone does not prove the cause. Shared-sample profile comparisons show change, not which profile is more accurate. Compare in 2D starting at L*=50, or switch to 3D. View gamut shows a separate ICC-derived surface; HTML can rotate it, PDF shows a still view.

The project already retains the ICC candidate. External saving creates a delivery copy with project name and date (YYMMDD); its ICC description is updated consistently, and original/delivery hashes distinguish the files. Export the complete HTML certificate folder with its supporting files. PDF can be handed over separately.

Certificates include project/material/coating/ink settings, instrument identity when available, FWA usage, ICC identity, scope, final measured colours, decision and dated signature space. Appendix A explains methods and ISO-related references; Appendix B contains legal terms. Selected ISO 12647-7:2016 comparison limits are referenced through MediaStandard Print 2018, published by Bundesverband Druck und Medien (bvdm, German printing and media industry association). This custom workflow is not ISO certification.

After row 17, run row 8. Then either perform current-iteration C2/C3 and approve for row 14, or use row 18 as decision support and row 19 for an explicitly numerical-only certificate. Earlier measurements remain historical evidence, not current-iteration verification. FWA models D50 effects from suitable M0 spectra; it does not turn an M0 instrument measurement into native M1. BPC (black point compensation) is not applied to the diagnostic absolute-colour comparison.

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Code, licensing and maintenance

InkProf code and authored documentation: GNU GPL v3 or later, without warranty; see LICENSE and THIRD_PARTY_NOTICES.md. Dependencies and schema/fonts have their own terms. This is a source-only release: no MATLAB, Python or Argyll runtime is included. See docs/releases/v1.1.0.md and licenses/review-v1.0.0.md for the checked release boundary and remaining validation limits.

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Shell refinement and continuation

Use Refinement tools to strengthen coverage near the gamut surface.

Review before printing

Choose the whole inner shell or selected areas, set geometric thickness in ΔE76, and review plotted candidates. Save the TIFF16 package and use its matching target for measurement. Saved packages are registered in step 15 without regenerating patch order.

Continue from measurements

Measure in step 16, build in 17 and compare in 18. The app suggests the next valid step. A stopped build retains measurements and the previous active ICC; inspect the latest C1/Grid reports from step 17 results before retrying.

Deliver with clear scope

Both print-verified and numerical-only certificates separate complete measurement records and technical details into measurement-appendix.pdf, linked by certificate ID and ICC hash. ColorChecker reference/measured pairs stay in the main certificate. Step 19 saves a numerical-only report and records the absence of separate print verification. External copies include the complete delivery bundle.